Concept 2 | Expected Value and Variance of Multiple Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$E(aX + b) = aE(X) + b$$

Variance

$$\mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X)$$

Independent

$$\mathrm{Var}(X + Y) = \mathrm{Var}\,X + \mathrm{Var}\,Y$$

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Classified Practice **Expected Value and Variance of Multiple Variables**

Q001 **Written**

Given the probability distributions of X and Y in the source tables, find E(X+Y).

Q002 **Written**

Three coins are tossed once; let X, Y and Z be the head indicators. Find E(X+Y+Z).

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Q003 Written

For the two source tables with X=1,2,3 and Y=4,5,6, find E(X+Y).

Q004 **Written**

For the two bags in the source exercise, find the expected total number of red balls drawn.

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Q005 **Written**

Three independent subsystems each receive two signals with success probability 0.5. Find E(X+Y+Z).

Q006 **Written**

Two 500-point coins and three 100-point coins are tossed. Find the expected total score Z.

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Q007 **Written**

Two 500-point coins and two 100-point coins are tossed. Find EZ.

Q008 **Written**

A die is thrown twice and the score is Z=3X+5Y. Find EZ.

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Expected Value and Variance of Multiple Variables

Q009 **Written**

For three 100-point coins and one 50-point coin, find the expected score from heads.

Q010 **Written**

In two independent raffles, buy two tickets from a 2/10 winner raffle and two from a 1/6 winner raffle. Find the expected prizes.

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Q011 **Written**

Three coins are tossed and a die is rolled; X is heads and Y is the die score. Find E(XY).

Q012 **Written**

For independent X with probabilities 2/5,2/5,1/5 at 1,2,3 and Y uniform on 2,4,6, find E(XY).

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Q013 **Written**

Ahmed tosses three coins and Mohamed tosses two coins. Let X and Y be their head counts. Find E(XY).

Q014 Written

A card numbered 1 to 4 is replaced and a second card is drawn. Find E(XY).

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Q015 Written

For independent X and Y with EX=1 and EY=2/7, find E(XY).

Q016 **Written**

A card X is drawn from 1,2,3 and three-coin heads Y is independent. Find V(X+Y) and sigma(X+Y).

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Q017 **Written**

Two independent dice are thrown. Find the variance and standard deviation of their sum.

Q018 **Written**

An independent coin head count X and die score Y are combined. Find V(X+Y) and sigma(X+Y).

Classified Practice **Expected Value and Variance of Multiple Variables**

Q019 **Written**

Ahmed tosses two coins and Mohamed tosses three coins. Find the variance of the total head count.

Q020 **Written**

Independent X is uniform on 1 to 4 and Y on 1 to 6. Find V(X+Y) and sigma(X+Y).

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Q021 Written

Three independent special dice take values 2,4,6 equally. Find E(XYZ) and V(X+Y+Z).

Q022 **Written**

Three independent dice take values 1,3,5 equally. Find E(XYZ) and V(X+Y+Z).

Classified Practice **Expected Value and Variance of Multiple Variables**

Q023 **Written**

Three ordinary dice are thrown. Find the expected product and the variance of their sum.

Q024 **Written**

Three friends have 5, 4 and 3 independent rolls to score a six. Find E(XYZ) and V(X+Y+Z).

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Q025 **Written**

If X, Y and Z are independent head counts from 4, 3 and 2 fair coins, find E(XYZ).

Q026 MCQ

E(X+Y) equals:

Choose the correct option.

A EX-EY

B EX+EY

C EXEY

D EX/EY

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Q027 **MCQ**

For three variables, E(X+Y+Z) equals:

Choose the correct option.

A $EXEYEZ$

B $EX+EY+EZ$

C $EX\text{-}EY\text{-}EZ$

D 0

Q028 MCQ

The expected heads in one fair coin toss is:

Choose the correct option.

A 0

B 1/2

C 1

D 2

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Q029 **MCQ**

For two signals with p=0.5, EX is:

Choose the correct option.

A 0.5

B 1

C 2

D 4

Q030 MCQ

If EX=2 and EY=5, E(X+Y) is:

Choose the correct option.

A 3

B 7

C 10

D 25

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Q031 *MCQ*

E(aX+bY) equals:

Choose the correct option.

A aEX+bEY

B abE(XY)

C EX+EY

D a+b

Q032 **MCQ**

The expected score of one fair 100-point coin is:

Choose the correct option.

A 25

B 50

C 100

D 200

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Q033 **MCQ**

For a fair die, $E(3X+5Y)$ with two rolls is:

Choose the correct option.

A 14

B 21

C 28

D 35

Q034 MCQ

A fixed entry fee changes:

Choose the correct option.

A variance only
B the mean only
C neither mean nor variance
D standard deviation only

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Q035 MCQ

The expected number of winners in 2 of 10 tickets with 2 winners is:

Choose the correct option.

A 1/5
B 2/5
C 1/2
D 4/5

Q036 MCQ

E(XY)=EXEY requires:

Choose the correct option.

A $X = Y$

B independence

C zero means

D equal variances

Classified Practice **Expected Value and Variance of Multiple Variables**

Q037 **MCQ**

For three fair coins, EX is:

Choose the correct option.

A 1/2

B 1

C 3/2

D 3

Q038 MCQ

For a fair die, EY is:

Choose the correct option.

A 2.5

B 3

C 3.5

D 6

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Q039 MCQ

With replacement, two card draws are:

Choose the correct option.

A dependent

B independent

C identical outcomes

D impossible

Q040 MCQ

If EX=2 and EY=3 for independent variables, E(XY) is:

Choose the correct option.

A 5

B 6

C 1

D 9

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Q041 **MCQ**

For independent X,Y, V(X+Y) equals:

Choose the correct option.

A $VX\text{-}VY$

B $VX+VY$

C $VXVY$

D 0

Q042 **MCQ**

The variance of a fair die is:

Choose the correct option.

A 7/2

B 35/12

C 35/6

D 1/4

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Q043 MCQ

Sigma(X+Y) is:

Choose the correct option.

A V(X+Y)

B sqrt$V(X+Y)$

C VX+VY

D E(X+Y)

Q044 MCQ

The variance of the sum of two independent fair dice is:

Choose the correct option.

A 35/12

B 35/6

C 70/6

D 49/4

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Expected Value and Variance of Multiple Variables

Q045 MCQ

For independent sums, a constant shift changes:

Choose the correct option.

A variance
B standard deviation
C neither spread measure
D both spread measures

Q046 MCQ

For independent X,Y,Z, E(XYZ) equals:

Choose the correct option.

A EX+EY+EZ

B EXEYEZ

C 0

D E(XY)+EZ

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Q047 MCQ

For independent X,Y,Z, V(X+Y+Z) equals:

Choose the correct option.

A $VXVYVZ$

B $VX+VY+VZ$

C $VX\text{-}VY\text{-}VZ$

D 0

Q048 MCQ

For a binomial count with n=5 and p=1/6, EX is:

Choose the correct option.

A 1/6

B 5/6

C 5/36

D 6

Classified Practice **Expected Value and Variance of Multiple Variables**

Q049 **MCQ**

For three values 2,4,6 equally likely, EX is:

Choose the correct option.

A 2
B 3
C 4
D 6

Q050 MCQ

For three independent fair dice, V(sum) is:

Choose the correct option.

A 35/12

B 35/6

C 35/4

D 343/8

Classified Practice **Expected Value and Variance of Multiple Variables**

Quiz WRITTEN 1 **Concept Quiz · Written**

If EX=4 and EY=7, find E(3X-2Y).

Quiz WRITTEN 2 **Concept Quiz · Written**

For independent variables with EX=2 and EY=5, find E(XY).

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Quiz WRITTEN 3 Concept Quiz · Written

Two independent fair dice are thrown. Find V(X+Y) and sigma(X+Y).

Quiz WRITTEN 4 **Concept Quiz · Written**

Three independent Binomial counts have (n,p)=(2,1/2),(3,1/2),(4,1/2). Find V(X+Y+Z).

Classified Practice **Expected Value and Variance of Multiple Variables**

Quiz MCQ 5 Concept Quiz · MCQ

If EX=3 and EY=4, E(2X+Y) is:

Choose the correct option.

A 7

B 10

C 14

D 24

Quiz MCQ 6 **Concept Quiz · MCQ**

For independent variables EX=2, EY=5, E(XY) is:

Choose the correct option.

A 7

B 10

C 25

D 1/2

Classified Practice **Expected Value and Variance of Multiple Variables**

Quiz MCQ 7 Concept Quiz · MCQ

For two independent die rolls, V(X+Y) is:

Choose the correct option.

A 35/12
B 35/6
C 35/4
D 7/2

Quiz MCQ 8 **Concept Quiz · MCQ**

For three independent variables, V(X+Y+Z) is:

Choose the correct option.

A the product of variances
B the sum of variances
C the sum of means
D always zero

Answer key

Use the question number shown on each page.

Q001 $EX = 35/3, EY = 9/2, so E(X + Y) = 97/6.$

Q002 $E(X + Y + Z) = 1/2 + 1/2 + 1/2 = 3/2.$

Q003 $E(X + Y) = EX + EY = 2 + 5 = 7.$

Q004 Bag A: 6/7; Bag B: 6/7; total $E = 12/7$.

Q005 Each mean is $2(0.5) = 1$, so $E(X + Y + Z) = 3$.

Q006 $Z = 500X + 100Y, so EZ = 5001 + 1003/2 = 650.$

Q007 $EZ = 5001 + 1001 = 600.$

Q008 $EZ = 37/2 + 57/2 = 28.$

Q009 $EZ = 350 + 25 = 175.$

Q010 $EZ = 22/10 + 21/6 = 11/15.$

Q011 X and Y are independent, so $E(XY) = EX$EY=3/27/2=21/4.

Q012 $EX = 8/5, EY = 4, so E(XY) = 32/5.$

Q013 $E(XY) = EXEY = 3/21 = 3/2.$

Q014 $EX = EY = 5/2, so E(XY) = 25/4.$

Q015 $E(XY) = EXEY = 2/7.$

Q016 VX=2/3, VY=3/4, so $V(X + Y) = 17/12$ and sigma=sqrt17/12.

Q017 $V(X + Y) = 35/12 + 35/12 = 35/6$; sigma=sqrt35/6.

Q018 $V = 1/4 + 35/12 = 19/6$; sigma=sqrt19/6.

Q019 $V = 21/4 + 31/4 = 5/4.$

Q020 $V = 5/4 + 35/12 = 25/6$; sigma=5/sqrt6.

Q021 EX=4, VX=8/3; $E(XYZ) = 64$ and $V(sum) = 8$.

Q022 EX=3, VX=8/3; $E(XYZ) = 27$ and $V(sum) = 8$.

Q023 $E(product)$ =7/$2^3 = 343/8$; $V(sum) = 335/12$=35/4.

Q024 Means are 5/6, 2/3, 1/2; $E(product) = 5/18$ and $V(sum) = 5/3$.

Q025 $E(XYZ) = 23/21$=3.

Q026 B

Q027 B

Q028 B

Q029 B

Q030 B

Q031 A

Q032 B

Q033 C

Q034 B

Q035 B

Q036 B

Q037 C

Q038 C

Q039 B

Q040 B

Q041 B

Q042 B

Q043 B

Q044 B

Q045 C

Q046 B

Q047 B

Q048 B

Q049 C

Q050 C

Quiz WRITTEN 1 −2

Quiz WRITTEN 2 10

Quiz WRITTEN 3 $V = 35/6$, sigma=sqrt35/6

Quiz WRITTEN 4 9/4

Quiz MCQ 5 B

Quiz MCQ 6 B

Quiz MCQ 7 B

Quiz MCQ 8 B